
Instructions

1. Write your Name and PID in the spaces provided above.
 2. Carefully read and complete the *Excel with Integrity Pledge* on the last page.
 3. Make sure your Name is on every page.
 4. No calculators, tablets, phones, or other electronic devices are allowed during this exam.
 5. Put away ANY devices that can be used for communication or can access the Internet.
 6. You may use one handwritten page of notes, but no books or other assistance during this exam.
 7. Read each question carefully and answer each question completely.
 8. Write your solutions clearly in the spaces provided.
 9. Show all of your work. No credit will be given for unsupported answers, even if correct.
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0. (1 point) Carefully read and complete the instructions at the top of this exam sheet and any additional instructions written on the chalkboard during the exam.



1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$L(x) = g(a) + g'(a)(x-a) = g(2) + g'(2)(x-2)$$

$$a=2 \quad L(x) = 5 - 2(x-2)$$

$$x=2.1 \quad L(2.1) = 5 - 2(2.1-2)$$

$$L(2.1) = 5 - 2(0.1)$$

$$L(2.1) = 5 - 0.2$$

$$L(2.1) = 4.8$$

$$\begin{array}{r} 2.1 \\ - 2.0 \\ \hline 0.1 \end{array}$$

$$\begin{array}{r} 0.1 \\ \cdot 2 \\ \hline 0.2 \end{array}$$

$$\begin{array}{r} 5.0 \\ - 0.2 \\ \hline 4.8 \end{array}$$

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(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

← derivative of $g(2)$

$$\lim_{x \rightarrow 2} \frac{5-5}{2-2} = 0$$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$5 - 2 \times 1.1$$

$$\begin{array}{r} 5.2 \\ - 0.2 \\ \hline 5.0 \end{array}$$

$$\begin{array}{r} 5.0 \\ - 0.2 \\ \hline 4.8 \end{array}$$

$$g(2) = 5$$

$$\lim_{h \rightarrow 0} \frac{g(2+h) - g(2)}{x+h-x} = g'(2) = -2$$

$$\lim_{h \rightarrow 0} \frac{g(2+h) - g(2)}{h} = \frac{g(2+h) - 5}{h}$$

$$g'(2) = \lim_{h \rightarrow 0} \frac{g(2+h) - g(2)}{h} = \lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2} = -2$$

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2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$\frac{1}{x} = x^{-1}$$

$$-x^{-1-1} = -x^{-2} = -\frac{1}{x^2}$$

$$f(x) = x^2 \sin\left(\frac{1}{x}\right)$$

$$m(x) = x^2 \rightarrow m'(x) = 2x$$

$$A(L) = \sin(L) \rightarrow A'(L) = \cos(L)$$

$$L(x) = \frac{1}{x} \rightarrow L'(x) = -\frac{1}{x^2}$$

$$k(x) = \sin\left(\frac{1}{x}\right) \rightarrow k'(x) = A'(L(x)) \cdot L'(x) = \cos\left(\frac{1}{x}\right) \cdot -\frac{1}{x^2}$$

$$f'(x) = m'(x)k(x) + m(x)k'(x)$$

$$2x \sin\left(\frac{1}{x}\right) + x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot -\frac{1}{x^2}$$

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 \sin\left(\frac{1}{x+h}\right) - f(0)}{h}$$

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 \sin\left(\frac{1}{x+h}\right) - 0}{h}$$

$$\lim_{h \rightarrow 0} \frac{x^2 \cdot \sin\left(\frac{1}{h}\right)}{h}$$

$$\lim_{h \rightarrow 0} h \sin\left(\frac{1}{h}\right) = 0$$

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h}$$

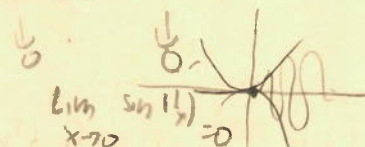
$$x^2 + 2xh + h^2 \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)$$

$$x^2 \frac{(2xh + h^2 \sin\left(\frac{1}{x+h}\right) - \sin\left(\frac{1}{x}\right))}{h}$$

Is it differentiable?

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$-x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$



Limit doesn't exist

because the function oscillates infinitely as x approaches 0.

Limit exists because the function oscillates within a bounded region as x approaches 0.

$$\lim_{h \rightarrow 0} \frac{h \sin\left(\frac{1}{h}\right)}{h} = 0$$



3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$s(t) = te^{kt}$$

$$\frac{d}{dx} e^{g(x)} = e^{g(x)} \cdot g'(x)$$

$$s'(t) = m'(t)A(t) + m(t) \cdot A'(t)$$

$$m(t) = t \rightarrow m'(t) = 1$$

$$A(t) = e^{kt} \rightarrow A'(t) = e^{kt} \cdot k$$

$$s'(t) = e^{kt} + tke^{kt}$$

$$s''(t) = p'(t) + L'(t) = ke^{kt} + tke^{kt} + ke^{kt}$$

$$p(t) = e^{kt} \rightarrow p'(t) = e^{kt} \cdot k$$

$$L(t) = tke^{kt} \rightarrow L'(t) = H(t)'(z(t)) + H(t)(z'(t)) = ke^{kt} + tke^{kt}$$

$$H(t) = t \rightarrow H'(t) = 1$$

$$z(t) = e^{kt} \rightarrow z'(t) = e^{kt} \cdot k$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$s''(t) = 2ke^{kt} + tke^{kt}$$

$$4 = 2ke^0 + 0$$

$$2 = k$$

$$s(t) = x e^{2t}$$

$$2xe^{2t} + e^{2t}$$

$$kte^{2t} + e^{2t}$$

$$ke^{2t} + kkte^{2t}$$

$$2ke^{2t} + kte^{2t}$$

$$s(t) = x \cdot e^{4t}$$

$$4 = 2ke^{kt} + k^2te^{kt}$$

$$s'(t) = e^{4t} + 4xe^{4t}$$

$$4 = 2ke^0 + 0$$

$$e^{kt} + kte^{kt}$$

$$4 = 2k$$

$$k = 2$$

$$s''(t) = 4e^{4t} + 4e^{4t} + 4x \cdot 4e^{4t}$$

$$4e^{4t} + 4e^{4t} + 4^2xe^{4t}$$

$$ke^{kt} + ke^{kt} + k^2te^{kt}$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$f(x) = \frac{1}{4}x^2 + \frac{1}{8}y^2 = 1$$

implicit differentiation

$$f'(x) = \frac{2}{4}x + \frac{2}{8}y \cdot y' = 0$$

$$\frac{2}{8}y \cdot y' = -\frac{1}{2}x$$

$$y' = -\frac{1}{2}x \cdot \frac{8}{2y}$$

$$y' = -\frac{8x}{4y} = -\frac{2x}{y}$$

input $(-\sqrt{2}, 2)$

$$y' = \frac{-2(-\sqrt{2})}{2} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

$$y = \sqrt{2}x + b \leftarrow \text{tangent line}$$

input $(-\sqrt{2}, 2)$

$$2 = \sqrt{2} \cdot -\sqrt{2} + b$$

$$2 = -2 + b$$

$$b = 4$$

$$L(x) = x\sqrt{2} + 4$$

$$\frac{2}{4}x + \frac{1}{4}y \cdot y' = 0$$

$$\frac{1}{4}y \cdot y' = -\frac{2}{4}x$$

$$y' = -\frac{1}{2}x \left(\frac{4}{y}\right)$$

$$= -\frac{4x}{2y} = -\frac{2x}{y}$$

check work

$$\frac{2\sqrt{2}}{2} = \sqrt{2}$$

$$y = \sqrt{2}x + b$$

$$2 = -\sqrt{2} \cdot \sqrt{2} + b$$

$$2 + 2 = b$$

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1. (6 points) A differentiable function $g(x)$ satisfies $g(3) = 7$ and $g'(3) = -3$.

(a) Estimate $g(3.1)$ using linear approximation.

$$m = -3$$

For when
 $x = 3$

$$y - y_1 = m(x - x_1) \quad y_1 = 7$$

$$x_1 = 3$$

$$y - 7 = -3(x - 3) \quad m = -3$$

$$y = -3x + 9 + 7$$

$$y = -3x + 16$$

For

$$g(3.1)$$

$$x = 3.1$$

$$y = -3(3.1) + 16 = -9.3 + 16$$

$$y = \boxed{6.7}$$

(b) Determine $\lim_{x \rightarrow 3} \frac{g(x) - 7}{x - 3}$.

$$\rightarrow \lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} = \lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a} \quad \leftarrow \text{definition}$$

$$g(3) = 7 \quad g'(3) = -3$$

~~Because~~ $\lim_{x \rightarrow 3} \frac{g(x) - 7}{x - 3}$ ~~seems like~~

$$a = 3$$

$$\text{So } \lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} = \frac{g(3)}{1} = g'(3)$$

$$g'(3) = \boxed{-3}$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$f(x) = x^2 \sin(\frac{1}{x})$$

$$f'(x) = uv' + u'v$$

$$u = x^2 \quad v = \sin(\frac{1}{x})$$

$$u' = 2x$$

$$v' =$$

$$f'(x) = x^2 \cdot \frac{1}{x^2} \cos(\frac{1}{x})$$

$$f'(x) = x^2 \cdot \left(-\frac{1}{x^2} \cos(\frac{1}{x}) \right) + 2x \cdot \sin(\frac{1}{x})$$

$$f'(x) = -\cos(\frac{1}{x}) + 2x \sin(\frac{1}{x})$$

$$z = \frac{1}{x} = x^{-1}$$

$$z' = -x^{-2} = -\frac{1}{x^2}$$

$$f(z) = \sin(z)$$

$$f'(z) = \cos(z)$$

$$\therefore \cos(\frac{1}{x}) \cdot \left(-\frac{1}{x^2} \right) = -\frac{1}{x^2} \cos(\frac{1}{x})$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

definition

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$f(x) = 0$$

when $x = 0$

Substitute h into $x^2 \sin(\frac{1}{x})$

then

$$f'(0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(\frac{1}{h}) - 0}{h}$$

$$f'(0) = \lim_{h \rightarrow 0} h \sin(\frac{1}{h})$$

Sin function exists between $-1 \leq \sin(\frac{1}{h}) \leq 1$ $\times$ by h

$\therefore -h \leq h \sin(\frac{1}{h}) \leq h$ as h approaches 0 so does $-h$

$\Rightarrow$ Squeeze Theorem $(\text{so } f'(0) = 0)$

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$s(t) = uv' + u'v$
 $u = t \quad v = e^{kt}$
 $u' = 1 \quad v' = ke^{kt}$
 $s'(t) = e^{kt} + te^{kt}$
 $s'(t) = e^{kt}(1+t)$

(b) Given that $s''(0) = 6$, determine the value of k .

Simplify

$$s'(t) = e^{kt}(1+t)$$

$$s''(t) = uv' + u'v$$

$$u = e^{kt} \quad v = 1+t$$

$$u' = ke^{kt} \quad v' = 1$$

$$s''(t) = (ke^{kt})(1+t) + (k)e^{kt}$$

$$= e^{kt}(k(1+t) + k)$$

$$= e^{kt}(k + k^2 + k)$$

$$s''(t) = e^{kt}(k^2 + 2k)$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(\sqrt{2}, 2)$.

$$\sqrt{\frac{x^2}{4}} + \sqrt{\frac{y^2}{8}}$$

$$8x^2 + 4y^2 = 32$$

$$\frac{d}{dx} 8x^2 + \frac{d}{dx} 4y^2 = \frac{d}{dx} 32$$

$$16x + 8y \frac{dy}{dx} = 0$$

$$y = \frac{16x}{8y}$$

$$\frac{dy}{dx} = -\frac{16x}{8y}$$

$$= -\frac{2x}{y}$$

Equation $x = \sqrt{2}$ $m = -\sqrt{2}$
 $y = 2$

Substitute
 $(\sqrt{2}, 2)$

$$\frac{dy}{dx} = m = -\frac{x(\sqrt{2})}{2}$$

$$m = -\sqrt{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\sqrt{2}(x - \sqrt{2})$$

$$y = -\sqrt{2}x + 2 + 2$$

$$y = -\sqrt{2}x + 4$$

$$y = -\sqrt{2}x + 4$$

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1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$L(x) = f(a) + f'(a)(x-a)$$

$$f(2) + f'(2)(2.1-2)$$

$$5 - 2(0.1)$$

$$3(0.1)$$

$$= 3 \times \frac{1}{10}$$

$$= 0.3, \frac{3}{10}$$

$$\frac{2.0}{\times 0.1}$$

$$\rightarrow 5 - 2(x-2)$$

$$5 - 2x + 4$$

$$x = 2.1$$

$$5 - 2(2.1) + 4$$

$$5 - 4.2 + 4$$

$$0.8 + 4$$

$$g(2.1) = 4.8$$

$$\begin{array}{r} 2.0 \\ \times 2.1 \\ \hline 20 \\ 40 \times \\ \hline 4.20 \end{array}$$

$$5 - 0.2$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

$$\lim_{x \rightarrow 2} \frac{g(2) - 5}{2 - 2} \quad \lim_{x \rightarrow 2} \frac{f(x) - f(a)}{x - a}$$

$$a = 2$$

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\frac{f(2+h) - f(2)}{h}$$

$$\frac{2+h-5}{h} = 2-5 = -3$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$\begin{aligned}
 f(x) &= x^2 \sin\left(\frac{1}{x}\right) && \text{product rule} \\
 u &= x^2 & v &= \sin\left(\frac{1}{x}\right) \\
 u' &= 2x & v' &= \cos\left(\frac{1}{x}\right) \cdot -\frac{1}{x^2} \\
 (2x) \left(\sin\left(\frac{1}{x}\right)\right) &+ (x^2) \left(-\frac{1}{x^2} \cos\left(\frac{1}{x}\right)\right) && \begin{aligned} x^{-1} & \rightarrow -x^{-2} \\ & \rightarrow -\frac{1}{x^2} \end{aligned} \\
 f'(x) &= 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) && \begin{aligned} x^2 \cdot -\frac{1}{x^2} & \rightarrow x^{-1} \\ -\frac{x^2}{x^2} & \rightarrow -x^{-2} \\ & \rightarrow -\frac{1}{x^2} \end{aligned}
 \end{aligned}$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$\begin{aligned}
 \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} & \quad x^2 \sin\left(\frac{1}{x}\right) \\
 \lim_{h \rightarrow 0} \frac{(x+h)^2 \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h} \\
 \frac{(x^2 + 2xh + h^2) \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h} \\
 \frac{x^2 + 2x + h \cdot \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h} \\
 \lim_{h \rightarrow 0} \frac{(x^2 + 2x) \sin\left(\frac{1}{x}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h} \\
 x^2 \sin\left(\frac{1}{x}\right) + 2x \sin\left(\frac{1}{x}\right) - x^2 \sin\left(\frac{1}{x}\right) \\
 x^2 \left(\sin\left(\frac{1}{x}\right) + 2x \sin\left(\frac{1}{x}\right) - \sin\left(\frac{1}{x}\right) \right)
 \end{aligned}$$

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$e^{kt} \cdot k$$

$$s(t) = te^{kt} \quad \text{- Product rule}$$

$$u = t \quad v = e^{kt}$$

$$u' = 1 \quad v' = ke^{kt}$$

$$1 \times e^{kt} + t \times ke^{kt}$$

$$s'(t) = e^{kt} + tke^{kt} \quad \text{or} \quad e^{kt}(1 + tk)$$

$$s''(t) = ke^{kt} + ke^{kt} + (t \cdot k^2 e^{kt})$$

$$2ke^{kt} + tk^2 e^{kt}$$

$$\text{or} \quad ke^{kt}(2 + tk)$$

$$\begin{matrix} t & k & e^{kt} \\ \downarrow & \downarrow & \\ u & v & \end{matrix}$$

$$u = t \quad v = ke^{kt}$$

$$u' = 1 \quad v' = k^2 e^{kt}$$

$$e^{2k} \cdot 2$$

$$ke^{kt} \cdot k$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$s''(0) = 2ke^{kt} + tk^2 e^{kt} = 4$$

$$2ke^0 + t \times 0 \times e^0 = 4$$

$$2k = 4$$

$$\boxed{k = 2}$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$y = m(x - x_1) + y_1$$

~~2x~~

~~dy~~
~~dx~~

$$\frac{d}{dx} \left(\frac{x^2}{4} + \frac{y^2}{8} \right) = \frac{d}{dx} [1]$$

$$\frac{2x}{4} + \frac{2y}{8} \cdot y' = 0$$

$$\frac{2x}{4} + \frac{y}{4} \cdot y' = 0$$

$$\frac{y}{4} \cdot y' = -\frac{2x}{4}$$

$$y' = \frac{-\frac{2x}{4}}{\frac{y}{4}}$$

$$\rightarrow \frac{-2x}{4} \times \frac{4}{y} = \frac{-2xy}{16}$$

$$\rightarrow \frac{-2x}{4} \times \frac{4}{y} = \frac{-2(-\sqrt{2})}{4} \times \frac{4}{2}$$

slope

↓

$$= \frac{2\sqrt{2}}{4} \times 2 = \frac{4\sqrt{2}}{4} = \sqrt{2}$$

$$y = \sqrt{2}(x + \sqrt{2}) + 2$$

$$y = \sqrt{2}x + 2 + 2$$

$$\boxed{y = \sqrt{2}x + 4}$$

$$y = m(x - x_1) + y_1$$

$$(-\sqrt{2}, 2) \quad \frac{-2x}{4} \times \frac{8}{y}$$

$$\frac{-2(-\sqrt{2})}{4} \times \frac{8}{2}$$

$$= \frac{2\sqrt{2}}{4} \times 4 = \frac{8\sqrt{2}}{4}$$

$$= 2\sqrt{2}$$

$$y = 2\sqrt{2}(x - \sqrt{2}) + 2$$

$$y = 2\sqrt{2}x - 2 + 2$$

$$y = 2\sqrt{2}x$$

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1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$L(x) = f(a) + f'(a) \cdot (x - a)$$

$$L(2.1) = 5 + (-2)(2.1 - 2)$$

$$= 5 + (-2)(.1)$$

$$= 5 - .2$$

$$= 4.8$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$. DNE

direct substitution: $\frac{5-5}{2-2} = \frac{0}{0}$ DNE

$$\lim_{x \rightarrow 2} \frac{(g'(x) \cdot (x-2)) - (g(x) - 5)}{(x-2)^2} = \lim_{x \rightarrow 2} \frac{g'(x) \cdot x - 2g'(x) - (g(x) - 5)}{x^2 - 4x + 4}$$

$$\frac{(-2 \cdot 2) - (2 \cdot -2) - 5 - 5}{2^2 - 4 \cdot 2 + 4} = \frac{-4 - 4 - 0}{4 - 8 + 4} = \frac{-8}{0} = \text{DNE}$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$f(x) = x^2 \sin\left(\frac{1}{x}\right) \quad \xrightarrow{x^{-1}}$$

$$\begin{aligned} f'(x) &= 2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right) \\ &= 2x \sin\left(\frac{1}{x}\right) + \frac{-x^2 \cos\left(\frac{1}{x}\right)}{x^2} \end{aligned}$$

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x^2 \sin\left(\frac{1}{x}\right) - 0 \sin\left(\frac{1}{0}\right)}{x} &\quad \text{DNE} \\ &= \text{Does not exist} \end{aligned}$$

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$\begin{aligned} s'(t) &= e^{kt} + te^{kt} \cdot k \\ &= e^{kt} + kt e^{kt} = e^{kt}(1+kt) = s'(t) \end{aligned}$$

$$\begin{aligned} s''(t) &= (e^{kt} \cdot k)(1+kt) + e^{kt} \cdot k \\ &= ke^{kt} + k^2 t e^{kt} + ke^{kt} \end{aligned}$$

$$= 2e^{kt} + k^2 t e^{kt}$$

$$s''(t) = e^{kt}(2 + k^2 t)$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$s''(t) = e^{kt} \cdot e^t \cdot (2 + k^2 t)$$

$$4 = 2e^{kt} + k^2 t e^{kt}$$

$$4 = e^k \cdot e^0 \cdot (2 + k^2 \cdot 0)$$

$$\ln 4 = 2 \ln kt + k^2 t \ln kt$$

$$4 = e^k(2)$$

$$\ln 4 - 2 \ln kt = k^2 t \ln kt$$

$$\ln 2 = e^k$$

$$\cancel{\ln 2 = k}$$

$$\frac{4}{2 + k^2 t} = e^{kt}$$

$$\frac{\ln 4 - 2 \ln kt}{k^2} = t \ln kt$$

$$\frac{4}{2} = e^{kt}$$

$$\frac{\ln 4 - 0}{k^2} = 0$$

$$k = 0.???$$

$$2 \neq 1$$

$$\frac{1}{k^2} = \text{DNE}$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$\frac{1}{4}x^2 + \frac{1}{8}y^2 = 1$$

$$\frac{2}{4}x + \frac{2}{8}yy' = 0$$

$$\frac{1}{2}x + \frac{1}{4}yy' = 0$$

$$\frac{1}{4}yy' = -\frac{1}{2}x$$

$$y' = \frac{-\frac{1}{2}x}{\frac{1}{4}y}$$

$$y' = -\frac{x}{\frac{1}{2} \cdot \frac{1}{y}}$$

$$y' = -\frac{2x}{y}$$

$$y' = \frac{-2(-\sqrt{2})}{2} = \sqrt{2}$$

$$y - 2 = \sqrt{2}(x + \sqrt{2})$$

$$y = \sqrt{2}x + 2 + 2$$

$$\boxed{y = \sqrt{2}x + 4}$$

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-

0. (1 point) Carefully read and complete the instructions at the top of this exam sheet and any additional instructions written on the chalkboard during the exam.

1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$5 + -2(2.1 - 2)$$

$$5 - .2$$

$$\boxed{4.8}$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

$$\frac{g'(x)}{1} \quad \frac{-2}{1}$$

$$\boxed{-2}$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$2x \left(\sin \frac{1}{x} \right) + \left(\cos \left(\frac{1}{x} \right) \right) \left(-\frac{1}{x^2} \right) (x^2)$$

$$\frac{-\cos\left(\frac{1}{x}\right) x^2}{x^2} - x^{-1} - x^2$$

$$\boxed{2x \left(\sin \frac{1}{x} \right) - \cos \left(\frac{1}{x} \right)}$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$\lim_{h \rightarrow 0} \left(\frac{x^2 + 2xh + h^2}{\sin(x+h)} - \frac{x^2}{\sin(x)} \right)$$

$$\lim_{h \rightarrow 0} \frac{(x+h)^2 \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h}$$

$$\lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2}{\sin(x+h)} - \frac{x^2}{\sin(x)}$$

$$x^2 \sin \frac{1}{x}$$

$$\boxed{f'(0) = 0}$$

$$x+h$$

$$\frac{(x^2 + 2xh + h^2) \sin\left(\frac{1}{x+h}\right) - x^2 \sin\left(\frac{1}{x}\right)}{h}$$

$$\frac{1}{\sin(x+h)}$$

$$x^2 + 2xh$$

$$x^2 \sin \frac{1}{x+h} + 2xh \sin \frac{1}{x+h} + (h^2) \sin \frac{1}{x+h} -$$

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$s'(t) = e^{kt} k$$

$$s''(t) = e^{kt} k$$

$$e^{kt} k$$

$$e^{kt} k$$

$$e^{kt} k$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$4 = e^{k \cdot 0} (k)$$

$$4 = k$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$\frac{2x}{4} + \frac{2yy'}{8}$$

$$\frac{x}{2} + \frac{yy'}{4} = 1$$

$$\frac{yy'}{4} = 1 - \frac{x}{2}$$

$$yy' = \left(1 - \frac{x}{2}\right)4$$

$$y' = \frac{\left(1 - \frac{x}{2}\right)4}{y}$$

$$\frac{\left(1 + \frac{\sqrt{2}}{2}\right)4}{2}$$

$$\left(1 + \frac{\sqrt{2}}{2}\right)2$$

$$2 + \frac{2\sqrt{2}}{2}$$

$$2 + \sqrt{2}$$

$$y = (2 + \sqrt{2})(x + \sqrt{2}) + 2$$

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0. (1 point) Carefully read and complete the instructions at the top of this exam sheet and any additional instructions written on the chalkboard during the exam.

1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$g'(2) = -2$$

$$g(2) = 5$$

$$L(x) = f(a) + f'(a)(x-a)$$

$$= 5 + (-2)(x-2)$$

$$= 5 - 2(x-2)$$

$$= 5 - 2x + 4$$

$$L(x) = \boxed{-2x + 9}$$

$$-2(2.1) + 9$$

$$= -1.1 + 9$$

$$= 7.9$$

$$L(x) = 5 - 2(2.1 - 2)$$

$$= \boxed{5 - 2(0.1)}$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

$$\lim_{x \rightarrow 2} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{x \rightarrow 2} \frac{(5+h) - 5}{h}$$

$$\lim_{x \rightarrow 2} \frac{-25 - 5h}{h}$$

$$\lim_{x \rightarrow 2} -$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$\begin{aligned} & \frac{d}{dx} \left[x^2 \sin\left(\frac{1}{x}\right) \right] \\ &= \frac{d}{dx} [x^2] \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \frac{d}{dx} \left[\sin\left(\frac{1}{x}\right) \right] \\ &= 2x \cdot \sin\left(\frac{1}{x}\right) + x^2 \cdot \cos\left(\frac{1}{x}\right) \cdot \frac{d}{dx} \left[\frac{1}{x} \right] \\ &= \boxed{2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \left(\frac{x-1}{x^2} \right)} \end{aligned}$$

$$\begin{aligned} & \frac{x \cdot \frac{d}{dx} [1] - 1 \cdot \frac{d}{dx} [x]}{x^2} \\ &= \frac{x \cdot 1 - 1 \cdot 1}{x^2} \\ &= \frac{x-1}{x^2} \end{aligned}$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$f(x) = 0$$

$$f(x+h) = 0$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x) = 0$$

$$\lim_{h \rightarrow 0} \frac{0 - 0}{h}$$

$$\lim_{h \rightarrow 0} \frac{0}{0}$$

Limit DNE

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$s'(t) = te^{kt} \rightarrow y = te^{kt}$$

$$\frac{d}{dx}[k] \cdot t + \frac{d}{dx}[t] \cdot k$$

$$\ln(y) = \ln(te^{kt})$$

$$\ln(y) = \ln(t) + \ln(e^{kt})$$

$$1 + t + 1 \cdot k$$

$$t + k$$

$$\frac{d}{dx} [1 + t + k(te^{kt})]$$

$$= \frac{d}{dx} [1 + t + k] \cdot te^{kt} + \frac{d}{dx} [te^{kt}]$$

$$\ln(y) = \ln(t) + kt \ln(e)$$

$$\frac{d}{dx} [\ln(y)] = \frac{d}{dx} [\ln(t)] + \frac{d}{dx} [kt]$$

$$= 2 \cdot te^{kt} + (1 + t + k)^2 (te^{kt})$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 1 + (t + k)$$

$$\frac{d}{dx} = (1 + t + k)(te^{kt})$$

(b) Given that $s''(0) = 4$, determine the value of k . $\rightarrow$ constant

$$t=4$$

$$= 2te^{kt} + (1 + t + k)^2 (te^{kt})$$

$$= 2(4)e^{k(4)} + (1 + 4 + k)^2 (4e^{k(4)})$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$\frac{d}{dx} \left[\frac{x^2}{4} \right] + \frac{d}{dx} \left[\frac{y^2}{8} \right] = \frac{d}{dx} [1]$$

$$\left(\frac{4 \cdot \frac{d}{dx} [x^2] - x^2 \cdot \frac{d}{dx} [4]}{4^2} \right) + \left(\frac{8 \cdot \frac{d}{dx} [y^2] - y^2 \cdot \frac{d}{dx} [8]}{8^2} \right) = 0$$

$$\left(\frac{(4 \cdot 2x) - (x^2 \cdot 0)}{16} \right) + \left(\frac{(8 \cdot 2y \cdot \frac{dy}{dx}) - y^2 \cdot 0}{64} \right) = 0$$

$$\left(\frac{8x}{16} \right) + \left(\frac{16y \frac{dy}{dx}}{64} \right) = 0$$

$$(64) \frac{16y \frac{dy}{dx}}{64} = -\frac{8x}{16} (64)$$

$$16y \frac{dy}{dx} = -\frac{8x(64)}{16}$$

$$\begin{array}{r} 3 \\ 64 \\ \times 8 \\ \hline 512 \end{array} \quad \begin{array}{r} 1 \\ 98 \\ + 3 \\ \hline 51 \end{array} \quad \begin{array}{r} 3 \\ 16 \\ \times 16 \\ \hline 96 \\ 160 \\ \hline 256 \end{array}$$

$$\frac{dy}{dx} = -\frac{8x(64)}{16} \cdot \frac{1}{16y}$$

$$\frac{dy}{dx} = -\frac{8x(64)}{16} \cdot \frac{1}{16y} =$$

$$= -\frac{512x}{16(16y)}$$

$$\frac{dy}{dx} = -\frac{512x}{256y}$$

$$\begin{aligned} m &= -\frac{512x}{256y} \\ &= -\frac{512(-\sqrt{2})}{256(2)} \end{aligned}$$

$$y - 2 = -\frac{512(-\sqrt{2})}{256(2)} (x - (-\sqrt{2}))$$

$$y = -\frac{512(-\sqrt{2})}{256(2)} (x + \sqrt{2}) + 2$$

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0. (1 point) Carefully read and complete the instructions at the top of this exam sheet and any additional instructions written on the chalkboard during the exam.

1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$\begin{aligned} g(2.1) &= -2(x + .1) + 5 \\ &= -2(2.1 + .1) + 5 \\ 5 - 4.4 &= \underline{0.6} \end{aligned}$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

$$\frac{5-5}{2-2} = \frac{0}{0} = \text{DNE}$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$2x \sin\left(\frac{1}{x}\right) - x^2 \cos\left(\frac{1}{x}\right)$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$f'(0)$ is 0 since the derivative of a sum of any single number is 0

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$s'(t) = e^{kt} + kte^{kt}$$

$$s''(t) = ke^{kt} + ke^{kt} + k^2 te^{kt}$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$4 = ke^{k0} + ke^{k0} + k^2 0 e^{k0}$$

$$4 = k + k + 0$$

$$k = 2$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$4x^2 + y^2 = 8$$

$$8x + 2yy' = 8$$

$$y' = \frac{8-8x}{2y}$$

$$y' = \frac{8+8(\sqrt{2})}{4}$$

$$y' = 2+2(\sqrt{2})$$

$$\frac{-\sqrt{2}^2}{4} + \frac{2^2}{8} = 1$$

$$-1/2 + 1/2 = 1$$

$$= 1$$

$$y = (2+2(\sqrt{2}))x + 1$$

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(a) Estimate $g(2.1)$ using linear approximation.

$$L(x) = f(a) + f'(a)(x-a)$$

$$L(2) = g(2) + g'(2)(x-2)$$

$$L(2) = 5 + (-2)(x-2)$$

$$L(2) = 3(x-2)$$

$$L(2) = 3x - 6$$

$$3(2.1) - 6 = \underline{0.3}$$

$$\begin{array}{r} 2.1 \\ \times 3 \\ \hline 6.3 \end{array}$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

~~$2.1 \times 3 = 6.3$~~

~~$g(x)$~~ $g(x) = g(2) = 5$

$$\frac{5-5}{2-2} = \frac{0}{0} = \underline{0}$$

~~$2.1 \times 3 = 6.3$~~

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$f(x) = x^2 \sin\left(\frac{1}{x}\right) \quad \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)$$

$$(2x) \left(\sin\left(\frac{1}{x}\right)\right) + (x^2) \left(\cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)\right)$$

$$2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

$$\boxed{2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)}$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (x+h)(x+h) = x^2 + 2hx + h^2$$

$$2(x+h) \frac{1}{\sin(x+h)} + (x+h)^2 \frac{1}{\cos(x+h)} \ln$$

$$\frac{2x+2h}{\sin x + \sinh} + \frac{x^2 + 2hx + h^2}{\cos x + \cosh}$$

$$\frac{2(x+h)}{\sin(x+h)} + \frac{(x+h)^2}{\cos(x+h)} \ln$$

$$\frac{(2x+2h)(\cos x + \cosh)}{(\sin x + \sinh)(\cos x + \cosh)} + \frac{(x^2 + 2hx + h^2)(\sin x + \sinh)}{(\cos x + \cosh)(\sin x + \sinh)}$$

$$\boxed{\frac{2x+2h+x^2+2hx+h^2}{(\cos x + \cosh)(\sin x + \sinh)}}$$

3. (6 points) An object moves along a line with position $s(t) = t e^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$t \cdot e^{kt}$$

product
rule

$$e^x \rightarrow \text{constant}$$

$$e^{kt}$$

$$(1+2t)$$

$$f: t \quad g: e^{kt}$$

$$f': 1 \quad g': e^{kt}$$

$$\frac{d}{dx} (t)(e^{kt}) + (t) \frac{d}{dx} (e^{kt})$$

$$(1)(e^{kt}) + (t)(e^{kt+k})$$

$$s'(t) = e^{kt} + (t)(e^{kt+k})$$

$$e^{kt} + (kt + t^2)e^{kt}$$

$$\frac{d}{dx} e^x = e^x \quad \frac{d}{dx} kt$$

product

$$f: kt \quad g: t$$

$$f': 1 \quad g': 1$$

$$y = e^{5x-2}$$

$$5x-2$$

$$ke^t$$

$$(kt+t^2)e^{kt}$$

$$s''(t) = (kt+t^2)e^{kt} + \frac{d}{dx} (kt+t^2)e^{kt}$$

$$(kt+t^2)(e^{kt}) + (e^{kt})(1+2t)$$

$$k^2t + kt^2 + kt^2 + t^3$$

$$t^3 + 2kt^2 + k^2t + e^{kt} + 2te^{kt} \quad \leftarrow s''(t)$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$2(4)^3 + 2k(4)^2 + k^2(4) + e^{k(4)} + 2(4)e^{k(4)}$$

$$\frac{16}{4}$$

$$64$$

$$64 + 32k + 4k^2 + e^{4k} + 8e^{4k}$$

$$64 + 32k + 4k^2 + 2e^{4k}$$

$$(0)^3 + 2k(0)^2 + k^2(0) + e^{k(0)} + 2(0)e^{k(0)}$$

$$0 + 0 + 0 + 1 + 0$$

$$\frac{64}{2} + \frac{32k}{2} + \frac{4k^2}{2} + \frac{2e^{4k}}{2}$$

$$k=2$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

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(a) Estimate $g(2.1)$ using linear approximation.

$$L(x) = f(a) + f'(a)(x-a)$$

$$L(2) = g(2) + g'(2)(x-2)$$

$$L(2) = 5 + (-2)(x-2)$$

$$L(2) = 3(x-2)$$

$$L(2) = 3x - 6$$

$$3(2.1) - 6 = \underline{0.3}$$

$$\begin{array}{r} 2.1 \\ \times 3 \\ \hline 6.3 \end{array}$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

~~$$\frac{g(2.1) - 5}{2.1 - 2} = \frac{0.3 - 5}{0.1} = \frac{-4.7}{0.1} = -47$$~~

~~$$g(x) = g(2) = 5$$~~

$$\frac{5-5}{2-2} = \frac{0}{0} = \underline{0}$$

~~$$\begin{array}{r} 2.1 \\ - 2.1 \\ \hline 0 \end{array}$$~~

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$f(x) = x^2 \sin\left(\frac{1}{x}\right) \quad \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$f'(x) = 2x \sin\left(\frac{1}{x}\right) + x^2 \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)$$

$$(2x) \left(\sin\left(\frac{1}{x}\right)\right) + (x^2) \left(\cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right)\right)$$

$$2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

$$\boxed{2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)}$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (x+h)(x+h) = x^2 + 2hx + h^2$$

$$2(x+h) \frac{1}{\sin(x+h)} + (x+h)^2 \frac{1}{\cos(x+h)} \ln$$

$$\frac{2x+2h}{\sin x + \sinh} + \frac{x^2 + 2hx + h^2}{\cos x + \cosh}$$

$$\frac{2(x+h)}{\sin(x+h)} + \frac{(x+h)^2}{\cos(x+h)} \ln$$

$$\frac{(2x+2h)(\cos x + \cosh)}{(\sin x + \sinh)(\cos x + \cosh)} + \frac{(x^2 + 2hx + h^2)(\sin x + \sinh)}{(\cos x + \cosh)(\sin x + \sinh)}$$

$$\boxed{\frac{2x+2h+x^2+2hx+h^2}{(\cos x + \cosh)(\sin x + \sinh)}}$$

3. (6 points) An object moves along a line with position $s(t) = t e^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$t \cdot e^{kt}$
product
rule

$e^x \rightarrow \text{constant}$
 e^{kt}

$$(1+2t)$$

$$f: t \quad g: e^{kt}$$

$$f': 1 \quad g': e^{kt}$$

$$\frac{d}{dx} (t)(e^{kt}) + (t) \frac{d}{dx} (e^{kt})$$

$$(1)(e^{kt}) + (t)(e^{kt+k})$$

$$s'(t) = e^{kt} + (t)(e^{kt+k})$$

$$e^{kt} + (kt + t^2)e^{kt}$$

$$s''(t) = (k+t)e^{kt} + \frac{d}{dx} (kt + t^2)e^{kt}$$

$$(kt + t^2)(k+t)e^{kt} + (e^{kt})(1+2t)$$

$$k^2t + kt^2 + kt^2 + t^3$$

$$t^3 + 2kt^2 + k^2t + e^{kt} + 2te^{kt} \quad \leftarrow s''(t)$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$2(4)^3 + 2k(4)^2 + k^2(4) + e^{k(4)} + 2(4)e^{k(4)}$$

$$\begin{array}{r} 16 \\ \times 4 \\ \hline 64 \end{array}$$

$$64 + 32k + 4k^2 + e^{4k} + 8e^{4k}$$

$$64 + 32k + 4k^2 + 2e^{4k}$$

$$(0)^3 + 2k(0)^2 + k^2(0) + e^{k(0)} + 2(0)e^{k(0)}$$

$$\frac{64}{2} + \frac{32k}{2} + \frac{4k^2}{2} + \frac{2e^{4k}}{2}$$

$$k=2$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

Instructions

1. Write your Name and PID in the spaces provided above.
 2. Carefully read and complete the *Excel with Integrity Pledge* on the last page.
 3. Make sure your Name is on every page.
 4. No calculators, tablets, phones, or other electronic devices are allowed during this exam.
 5. Put away ANY devices that can be used for communication or can access the Internet.
 6. You may use one handwritten page of notes, but no books or other assistance during this exam.
 7. Read each question carefully and answer each question completely.
 8. Write your solutions clearly in the spaces provided.
 9. Show all of your work. No credit will be given for unsupported answers, even if correct.
-
0. (1 point) Carefully read and complete the instructions at the top of this exam sheet and any additional instructions written on the chalkboard during the exam.

1. (6 points) A differentiable function $g(x)$ satisfies $g(2) = 5$ and $g'(2) = -2$.

(a) Estimate $g(2.1)$ using linear approximation.

$$L(x) = f(a) + f'(a)(x-a)$$

$$5 = g(2) + g'(2)(x-2)$$

$$5 = g(2) + -2(x-2)$$

$$5 = 5 + -2(x-2)$$

$$5 = 5 - 2x + 2x + 4 \quad \cdot 1 = x$$

$$\underline{-5 = -4x + 9} = \frac{-4}{-4} = \frac{-4x}{4}$$

(b) Determine $\lim_{x \rightarrow 2} \frac{g(x) - 5}{x - 2}$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 2} \frac{f(2) - 5}{2 - 2}$$

2. (6 points) Consider the function $f(x) = \begin{cases} 0 & \text{if } x = 0, \\ x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0. \end{cases}$

(a) Compute $f'(x)$ for $x \neq 0$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx} \left[x^2 \sin\left(\frac{1}{x}\right) \right]$$

(b) Compute $f'(0)$ using the limit definition of derivative. Be sure to justify your answer.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

3. (6 points) An object moves along a line with position $s(t) = te^{kt}$.

(a) Compute $s'(t)$ and $s''(t)$.

$$s(t) = te^{kt}$$

$$v(t) = s'(t)$$

$$a(t) = v'(t) = s''(t)$$

$$v(t) = te^{kt}$$

$$a(t) = kt^2e^{kt} = s''(t)$$

$$s(t) = kt^2e^{kt}$$

(b) Given that $s''(0) = 4$, determine the value of k .

$$s''(0) = 4$$

4. (6 points) Find equation for the line tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{8} = 1$ at the point $(-\sqrt{2}, 2)$.

$$\frac{x^2}{4} + \frac{y^2}{8} = 1 \quad (-\sqrt{2}, 2)$$

$$n = \frac{d}{dx} \left[\frac{x^2}{4} \right] + \frac{d}{dx} \left[\frac{y^2}{8} \right]$$

$$\frac{d}{dx} \left[\frac{(-\sqrt{2})^2}{4} \right] + \frac{d}{dx} \left[\frac{2^2}{8} \right]$$

$$\frac{d}{dx} \left[\frac{-4}{4} \right] + \frac{d}{dx} \left[\frac{4}{8} \right]$$

$$\frac{d}{dx} [-1] + \frac{d}{dx} \left[\frac{1}{2} \right]$$

$$0 + 0$$

$$y = 0 + \Delta \text{ on } \frac{1 \Delta}{h}$$

$$y = 0 + (1 - 1) + 0$$